

L'Hôpital's Rule: the $\frac{*}{\infty}$ Form

The usual L'Hôpital's rule handles $\frac{0}{0}$ and $\frac{\infty}{\infty}$. In many problems, however, we do not know what the numerator does—it may be bounded, oscillating, or unknown—while the denominator diverges. The version below covers exactly this case: we need no condition on $f(x)$ at all.

Theorem (L'Hôpital, $\frac{*}{\infty}$ form). *Let $a \in \mathbb{R} \cup \{-\infty\}$ and let f, g be differentiable on (a, ∞) . Suppose*

1. $\lim_{x \rightarrow \infty} g(x) = \infty$,
2. $g'(x) \neq 0$ on (a, ∞) ,
3. $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = L$ for some $L \in \mathbb{R}$.

Then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L.$$

Proof. Fix $\varepsilon > 0$. Since $f'/g' \rightarrow L$, choose $y > a$ such that

$$L - \varepsilon < \frac{f'(x)}{g'(x)} < L + \varepsilon \quad \text{for all } x > y.$$

Fix such a y for the rest of the argument. For any $x > y$, Cauchy's Mean Value Theorem gives some $c \in (y, x)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(c)}{g'(c)}.$$

Since $c > y$, we get

$$L - \varepsilon < \frac{f(x) - f(y)}{g(x) - g(y)} < L + \varepsilon.$$

As $g(x) \rightarrow \infty$, for x large enough we have $g(x) > g(y)$, so $g(x) - g(y) > 0$. Multiplying through,

$$(L - \varepsilon)(g(x) - g(y)) < f(x) - f(y) < (L + \varepsilon)(g(x) - g(y)).$$

Adding $f(y)$ and dividing by $g(x) > 0$,

$$\frac{f(y)}{g(x)} + (L - \varepsilon) \left(1 - \frac{g(y)}{g(x)}\right) < \frac{f(x)}{g(x)} < \frac{f(y)}{g(x)} + (L + \varepsilon) \left(1 - \frac{g(y)}{g(x)}\right).$$

Here y is fixed, so $f(y)$ and $g(y)$ are constants. Letting $x \rightarrow \infty$,

$$\frac{f(y)}{g(x)} \rightarrow 0, \quad \frac{g(y)}{g(x)} \rightarrow 0.$$

Taking $\liminf$ and $\limsup$ of the middle term,

$$L - \varepsilon \leq \liminf_{x \rightarrow \infty} \frac{f(x)}{g(x)} \leq \limsup_{x \rightarrow \infty} \frac{f(x)}{g(x)} \leq L + \varepsilon.$$

This holds for every $\varepsilon > 0$, so both equal L . Hence $\lim_{x \rightarrow \infty} f(x)/g(x) = L$. □

Remark. The proof never uses anything about $\lim_{x \rightarrow \infty} f(x)$. Fixing y locks in the bounds on f'/g' ; then sending $x \rightarrow \infty$ makes the $f(y)/g(x)$ and $g(y)/g(x)$ terms vanish because $g(x) \rightarrow \infty$. Whatever f is doing gets absorbed by the denominator.